Université d'Ottawa · University of Ottawa

Faculté des sciences
Mathématiques et de statistique

Faculty of Science
Mathematics and Statistics

MAT 2143 – The Final Exam **Instructor: K. Zaynullin**

Last name: _____

First name: _____

Student number:

Please, read the following instructions carefully:

- You have 3 hours to complete this exam. Read each question carefully. Where it is possible to check your work, do so.
- This is a closed book exam, and no notes of any kind are allowed. The use of calculators, cell phones, pagers or any text storage or communication device is not permitted.
- Answer all questions in the space provided after each question. Each answer requires justification written legibly and logically: you must convince the marker that you know why your solution is correct. Use the backs of pages if necessary.

THIS SPACE IS RESERVED FOR THE MARKER:

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Total
Mark															
Out of	3	2	2	3	3	3	3	2	3	3	4	3	3	3	40

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1. Determine whether or not the relation

$$x \sim y \quad \text{if } |x - y| \leq 4$$

is an equivalence relation on the set of real numbers $\mathbb{R}$.

It is an equivalence relation (Y/N): ☐ (1)

Justification: (2)

2. Using the induction prove that $n! > 2^n$ for all $n \geq 4$.

(here $n! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot n$)

Justification: (2)

3. Find all solutions of the following system of congruences

$$x \equiv 2 \pmod 5$$
$$x \equiv 6 \pmod{11}$$

ANSWER: (1)

Justification: (1)

4. Given the groups $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{Z}, +)$, let

$$G = (\mathbb{R} \setminus \{0\}) \times \mathbb{Z} \quad \text{(the Cartesian product of sets).}$$

Define a binary operation '$\circ$' on G by

$$(a, m) \circ (b, n) = (a \cdot b, m + n), \qquad a, b \in \mathbb{R} \setminus \{0\}, \quad m, n \in \mathbb{Z}.$$

Show that G is a group under this operation.

Justification: (3)

5. Describe (draw the respective Lattice Diagram) all the subgroups of the (quaternion) group

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k \mid i^2 = j^2 = k^2 = ijk = -1\}.$$

Lattice Diagram: (3)

6. Let p and q be distinct primes. How many generators does $\mathbb{Z}/pq\mathbb{Z}$ (the group of integers modulo pq with respect to addition) have?

ANSWER: [ ] (a formula in terms of p and q) (1)

Justification: (2)

7. Consider the group of invertible elements $G = (\mathbb{Z}/n\mathbb{Z})^\times$, $n > 2$ (with respect to the multiplication).

Prove that there exists an element $g \in G$ such that $g^2 = 1$ and $g \neq 1$.

Justification: (3)

8. (a) Compute

$$(1254)^{-1}(123)(45)(1254)$$

ANSWER: (1)

(b) Express the following permutation as product of transpositions

$$(1426)(142)$$

ANSWER: (1)

9. Does the group of even permutations A_8 contain an element of order 26?

ANSWER (Y/N): (1)

Justification: (2)

10. Describe all group homomorphisms from $(\mathbb{Z}/24\mathbb{Z}, +)$ to $(\mathbb{Z}/18\mathbb{Z}, +)$.

Answer/Justification: (3)

11. Let $Z(G)$ be the centre of G. Prove that if the factor group $G/Z(G)$ is cyclic, then G is abelian.

Justification: (4)

12. Let R be a ring. Define the centre of R to be

$$Z(R) = \{a \in R \mid \ ar = ra \text{ for all } r \in R \ \}.$$

Prove that $Z(R)$ is a commutative subring of R.

Justification: (3)

13. Prove that $(\mathbb{R}, +, \cdot)$ is not isomorphic to $(\mathbb{C}, +, \cdot)$ as a ring.

Justification: (3)

14. Let R be a commutative ring. Show that the set of all nilpotent elements in R forms an ideal.

Justification: (3)